Daniel Aarao Reis Arturi

AI Safety and Agent Security Researcher

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RESEARCH PROFILE

AI-safety researcher and agent-systems engineer who builds reproducible evaluation harnesses, instruments long-horizon agent behavior, and publishes empirical findings. Experience spans emergent misalignment, intervention evaluation, multi-agent orchestration, persistent memory, tool integration, and auditable human-in-the-loop systems.

Research interests AI agent security, adversarial-behavior evaluation, agent guardrails, memory and context integrity, mechanistic interpretability, threat modeling, and reproducible evaluation

RELEVANT RESEARCH EXPERIENCE

Algoverse AI Research — Apr 2025-Present

Research Intern | AI Safety and Mechanistic Interpretability | Paper | Project | Code

- Studied emergent misalignment in LoRA-adapted Qwen2.5-7B and Llama 3.1-8B models across narrowly harmful tasks, finding shared parameter subspaces with mean principal angles near 20 degrees versus an approximately 85-degree random baseline and projection overlap near 0.8.
- Built reproducible fine-tuning and behavioral evaluation pipelines, then used linear mode connectivity to test functional equivalence across independently trained models; cross-task interpolations retained misaligned behavior while preserving more than 90% coherence.
- Extended the work to post-hoc and during-training interventions, measuring mitigation alongside GSM8K capability retention, and began a separate study of how multi-thousand-turn conversations and persistent memory affect model persona stability.
- Co-authored the resulting NeurIPS 2025 workshop paper as a joint first author; the work received a UniReps oral award, a MechInterp honourable mention, and a CogInterp poster acceptance.

NYU Tandon School of Engineering, VIDA Center — May 2026-Present

Research Scientist | DARPA expMath | Lea | Program

- Helped develop Lea, an open-source agent platform for mathematician-controlled formalization with proof decomposition, bounded sub-agent roles, mid-run intervention, trajectory inspection, manual correction, and auditable Lean 4 output.
- Worked on the project-aware agent harness and its research infrastructure, integrating tool and skill access, MCP servers, Lean checking, typed event streams, persistent proof history, and shared state across standalone and Overleaf clients.
- Developed review and recovery workflows around agent failures, including dependency-aware execution, counterexample handling, theorem-specific review, self-repair, resumable sessions, and GitHub export.
- Contributed to long-horizon agent evaluations, related-work synthesis, architecture documentation, case-study analysis, and the human-in-the-loop research framing; GitHub listed me as Lea's leading public contributor with 144 contributions in September 2026.

SELECTED SECURITY-MINDED SYSTEM

NYC Housing Research

Python, FastAPI, SQLite, retrieval and evaluation [Code]

- Built a local-first cited research system over official NYC legal sources with source provenance, explicit coverage gaps, report-bound human review, citation-support checks, and controls separating keyless retrieval from optional semantic indexing and generation.
- Implemented dual authentication paths, row-level access controls, provider-cost approvals, backup and recovery workflows, and automated regression coverage spanning ingestion, retrieval, security, migration, desktop, and browser behavior.

PUBLICATIONS AND MANUSCRIPTS

Daniel Aarao Reis Arturi*, Eric Zhang*, Andrew Ansah, Kevin Zhu, Ashwinee Panda, and Aishwarya Balwani. *Shared Parameter Subspaces and Cross-Task Linearity in Emergently Misaligned Behavior*. arXiv:2511.02022, 2025. Joint first author. [Project] [Code]

* Joint first authors with equal contributions; authors listed alphabetically. Expanded manuscript: *Of Basins and Vaccines: Emergent Misalignment Through the Lens of Parametric Geometry* (ICLR submission).

Daniel Aarao Reis Arturi, Christoph Köhnen, George Fletcher, Bettina Kemme, and Stefanie Scherzinger. *Seeing the Trees for the Forest: Leveraging Tree-Shaped Substructures in Property Graphs*. arXiv:2603.12476, 2026. Under review at EDBT 2027. [Artifact]

Daniel Aarao Reis Arturi, Emilie Fahber, Stefanie Scherzinger, and Bettina Kemme. *Evaluating a Quantum Solution: How hard can it be?*. 2nd Workshop on Quantum Computing and Data/Knowledge Management, VLDB 2026. First author and oral presenter. [Artifact]

EDUCATION

New York University, Courant Institute of Mathematical Sciences | M.S. Computer Science | Fall 2026-Present

McGill University | B.Sc. Honours Software Engineering | May 2026 | GPA 3.99/4.0 | First Class Honours

RESEARCH METHODS AND TECHNICAL SKILLS

AI safety and evaluation Adversarial-behavior evaluation, intervention evaluation, long-horizon agent analysis, mechanistic interpretability, parameter-subspace analysis, linear mode connectivity, exact-oracle validation, statistical analysis, technical writing

Agents and ML Python, PyTorch, Transformers, LoRA/PEFT, NumPy, pandas, scikit-learn, agent orchestration, MCP integration, persistent memory, tool-use instrumentation, Jupyter, CUDA, Slurm

Systems Lean 4, Mathlib, FastAPI, React, TypeScript, Node.js, SQLite, PostgreSQL, Docker, Git, GitHub Actions, Linux